Comparison of Utility Bias for Context Attribution

July 29, 2026

1 Real-World Data

1.1 Ranking Disagreement

When we rank context sentences by their attribution value, we produce 4 sets of rankings for each dataset: with 2 methods (ContextCite and Exact Shapley) and 2 types of utility function (similarity and likelihood).

Table 1: Comparison of rankings between ContextCite and Exact Shapley.

	2WikiQA		**HotPotQA**		**MusiQue**	
Utility Function	ρ	RBO	ρ	RBO	ρ	RBO
Likelihood	0.8328	0.9139	0.8397	0.9458	0.8454	0.9347
Similarity	0.7791	0.8526	0.7667	0.8329	0.7685	0.8696

We evaluate ranking fidelity across utility functions to validate ContextCite as a scalable approximation for Exact-Shapley. Table 1 demonstrates that ContextCite maintains a strong rank correlation across datasets. Because identifying top evidence is critical, we applied Rank-Biased Overlap (RBO, p=0.5), which yielded even higher and consistent fidelity scores. High RBO scores confirm that ContextCite accurately preserves the highest-ranked documents, establishing it as a reliable proxy for exact Shapley calculations.

Having validated ContextCite, we compare utility functions (Table 2). Consistently low scores confirm our hypothesis: syntactic and semantic attributions fundamentally diverge. However, a moderate Spearman's ρ suggests these rankings are penalized by the random sorting of irrelevant tail documents. By applying Rank-Biased Overlap (RBO, $p = 0.5$) to isolate the most critical evidence, we see higher agreement. This indicates that while Likelihood and Similarity functions agree on the primary ground-truth fact (Rank 1), their distinct biases

Table 2: Comparison of rankings between semantic and likelihood utility functions.

	2WikiQA				HotPotQA				MusiQue			
Method	ρ	RBO	R1%	R2%	ρ	RBO	R1%	R2%	ρ	RBO	R1%	R2%
Exact Shapley	0.447	0.595	54	30	0.520	0.678	74	48	0.581	0.744	74	42
ContextCite	0.448	0.620	60	32	0.582	0.678	70	38	0.572	0.735	74	48

(lexical priming vs. topic alignment) cause immediate divergence on secondary supporting evidence.

To understand the practical implications of this divergence, we evaluate Rank1 (for the top-ranked document) and Rank2 (for the top two documents) agreement across all queries (these scores are denoted as R1 and R2 in Table 2. Likelihood and Similarity functions tend to identify the same primary document, though the magnitude differs across the datasets. Thus, the heavy top-weighted RBO scores indicate that outside of this primary anchor point, utility functions' biases cause their rankings for the remaining supporting evidence to collapse in agreement.

1.2 Ground truth disagreement

To find out which utility-type better agrees with ground truth, we calculate Precision, Recall, F1-Score and NDCG. We apply distinct rank cutoffs tailored to the structural constraints of our datasets. Because the supporting documents of our multi-hop queries consist of exactly two necessary premises, we evaluate classification metrics (Precision, Recall, and F1-Score) strictly at $K = 2$. Conversely, we measure positional ranking quality using NDCG@3; expanding this cutoff by one allows the metric to explicitly capture and penalize the rank intrusion of distractors. The results are shown in Table 3. Likelihood generally outperforms similarity across both retrieval and ranking metrics. It demonstrates that similarity-based attribution systematically falls victim to "alignment without sufficiency".

A more detailed picture for the 2WikiMultihopQA dataset with various K values (specifically, $K \in [1, 3, 5]$) is shown in Figure 1.

1.3 Illustrative examples

Although aggregate metrics highlight a performance gap between the two utility types, they obscure the specific mechanisms driving this behavior. To understand this divergence, we present the following document-level case studies which illustrate how similarity systematically degrades by falling for "topic

Table 3: Ground truth evaluation metrics across datasets and methods.

Metric	Utility	2WikiQA		HotPotQA		MusiQue	
		Ex-Sh	CC	Ex-Sh	CC	Ex-Sh	CC
NDCG@3	Likelihood	0.8872	0.9072	0.9474	0.9363	0.9435	0.9422
	Similarity	0.7517	0.7617	0.8848	0.8645	0.9403	0.9315
Precision@2	Likelihood	0.6067	0.6067	0.6000	0.5933	0.6000	0.6067
	Similarity	0.4933	0.4933	0.5267	0.4667	0.5800	0.5533
Recall@2	Likelihood	0.9100	0.9100	0.9000	0.8900	0.9000	0.9100
	Similarity	0.7400	0.7400	0.7900	0.7000	0.8700	0.8300
F1-Score@2	Likelihood	0.7280	0.7280	0.7200	0.7120	0.7200	0.7280
	Similarity	0.5920	0.5920	0.6320	0.5600	0.6960	0.6640

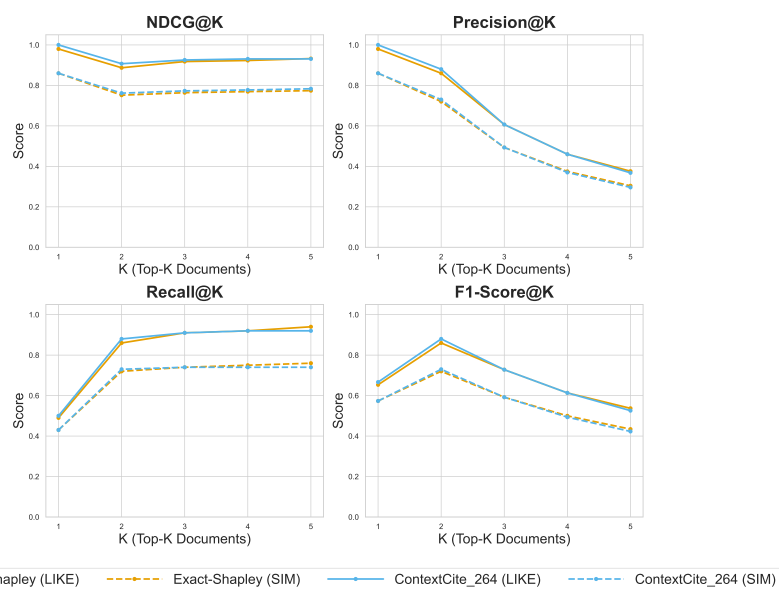
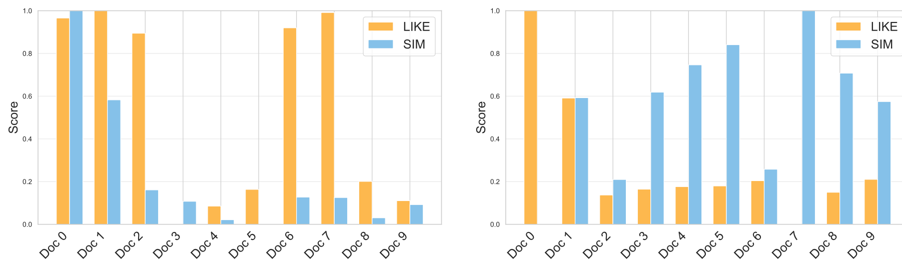


Figure 1: NDCG, Recall, Precision, F1 performance at Top-K across methods and utility functions for 2WikiMultihopQA.

traps" that promote aligned distractors over necessary facts. In contrast, while likelihood reliably maintains logical sufficiency, it remains uniquely vulnerable to exact-match "lexical traps".



(a) Similarity works better; Likelihood tricked by syntactic bias.

(b) Likelihood works better; Similarity tricked by semantic bias.

Figure 2: Document-level utility comparison comparing Similarity and Likelihood.

Figure 2a illustrates the case where similarity correctly ranks the documents while likelihood does not. The QA specifics and the top-ranked documents are presented in Table 4. Likelihood in this example acts like a basic keyword search and ranks highly documents that mention "regulations". When it sees those exact tokens in the context window, it calculates that they increase the raw probability of the generated text, and scores them highly.

Table 4: QA specifics and document utilities for the Hacker-Pschorr example.

Question	What does the Hacker-Pschorr Brewery have to limit in order to comply with German regulations?
Answer	ingredients in beer
Top 2 by LIKE	**Doc 1:** The Reinheitsgebot ... is the collective name for a series of regulations limiting the ingredients in beer in Germany. **Doc 7:** Code of Federal Regulations, Title 47, Part 15 (47 CFR 15) is an oft-quoted part of Federal Communications Commission (FCC) rules and regulations regarding unlicensed transmissions.
Top 2 by SIM	**Doc 0:** Hacker-Pschorr is a brewery in Munich, formed in 1972 out of the merger of two breweries, Hacker and Pschorr. **Doc 1:** The Reinheitsgebot ... is the collective name for a series of regulations limiting the ingredients in beer in Germany.

Figure 2b illustrates the contrary case where the likelihood correctly ranks the documents while the semantic one is spread uniformly across different documents. The top two selected documents for each utility are shown in Table 5. In this example, we can see that the similarity is heavily distracted by the noise in the documents, all of which speak about country singers, songs, and duets.

Table 5: QA specifics and document utilities for the Tammy Wynette example.

Question	What American country music singer-songwriter, born in May of 1942, sang a duet with her ex-husband the same year that he released the song “The Battle?”
Answer	Tammy Wynette
Top 2 by LIKE	**Doc 0:** “The Battle” is a song by American country music artist George Jones. ... Meanwhile, the single "Golden Ring," a duet with his estranged ex-wife Tammy Wynette, became a #1 hit that same year. **Doc 1:** Tammy Wynette ... was an American country music singer-songwriter and one of country music’s best-known artists and biggest-selling female singers.
Top 2 by SIM	**Doc 5:** “Three Wooden Crosses” the title of a song written by Kim Williams and Doug Johnson, and recorded by American country music singer-songwriter Randy Travis. ... **Doc 7:** “Please Don’t Stop Loving Me” is a song written and recorded as a duet by American country music artists Porter Wagoner and Dolly Parton. ...